introduction C programming language

error

- Semantic error
- Syntax error

Debugger

bug detection

editor

Compiler

Debugger

} IDE integrated Development environment

System programming language

Unix Os

1972

Mid-level programming language

Case-Sensitive

Compiler (Derleyici)

Source file(s) (text)

Object file(s) (binary)

Linker

Executable File (Binary)

IDEs

Dev C++

Visual Stdio

Eclipse

Netbeans

CodeBlocks

Clion

toplona.c (source file)

toplona.o (object file)

toplona / toplona.exe (executable File)

↳ Windows

↳ linux
macOS

```
#include <stdio.h>          " stdio.h "

int main(){

return 0;  ——————→  exit(0);

}
```

Read / write operations

scanf("%d", &o); gets()

↳ Variable

→ ampersand

puts()

```
printf("%d", a);
printf("Hello world");
printf("Hello world \n");
scanf("%d %d", &a, &b);
scanf("%d", &a);
scanf("%d", &b);
printf("A'nın değeri : %d - B'nin değeri
        %d", a, b);
```

Aritmetic operations

\+ addition

\- subtraction

* multiplication

/ Division

% modular arithmetic

5/2 → 2 Div

5.0 / 2
5 / 2.0
5.0 / 2.0
} 2.5

Control flow (if)

if(x) → non-zero (true)
→ zero (false)

statement 1,
statement 2,

if(1)
if(5)
if(-5)
} true

if(0) → false

```
if(x) {
statement 1;
}
else {
```

```
statement 2;
}
statement 3;
```

```
if(x) {
  statement 1;
  statement 2;
}
else {
  statement 3;
  statement 4;
}
statement 5;
```

Comparision expressions

`<` less than

`>` greater than

`<=` less than or equal to

`>=` greater than or equal to

`==` equal to

!= not equal to

```
if(-1<0){
}
else {
}

if(0<-1){
}
else {
  :
}

if(-1<0<1){
    1<1
     ∅
}
else {
  :
}

if(-1<∅<2){
```

1

}

else {

}

logical operations

And &&

Or ||

x < y < z

if ((x < y) && (y < z))

if ((x < 0) || (x > 100))

Nested if statements

```
if(a<b){
    if(a<c){
        d=a;
    }
    else{
        d=c;
    }
}
```

a, b, c

3 4 5

```
}
else if(b<c){
    d=b
}
else {
  d=c
}
printf("%d", d);
```

Scalor Data types

Char (1 byte)
int
floot
double

Short unsigned
long signed

char < short int ≤ int ≤ long int

floot < double

short int (2byte)
long int 4-8 byte
int 4 byte

unsigned int a;

Variable declaration

```
int a,b,c;

float x; x=0.0;
```

```
double y,z ;
char harf,m;
float x=0.0;

int main(){
int num1, num2, result;
printf("iki sayi giriniz\n");
scanf("%d", &num1);
scanf("%d", &num2);
result = num1+num2;
printf("sonuc = %d", result);
return 0;
}
```

↳ iki sayi giriniz

5 (enter)

6 (enter)

Sonuç → 11

Loops

for
while
do while

for statement

```
for(i=0;i<100;i++){
```

exp1 → `i=0`
exp2 → `i<100`
exp3 → `i++`

```
    statement 1;
```

`i=i+2` `i=i*2`

INC	ADD
i++ ↔	i=i+1
Dec	sub
i--	i=i-1

Do-while statement

i=0

```
do {

i=i+1
statement1;

} while(i<100);
```

While statement

```
i=0
while(i<100){
    statement1;
     i++;

}

scanf("%d %d", &x, &y)

 while(x<y){

       x++;

     y--;
   }
```

Arrays

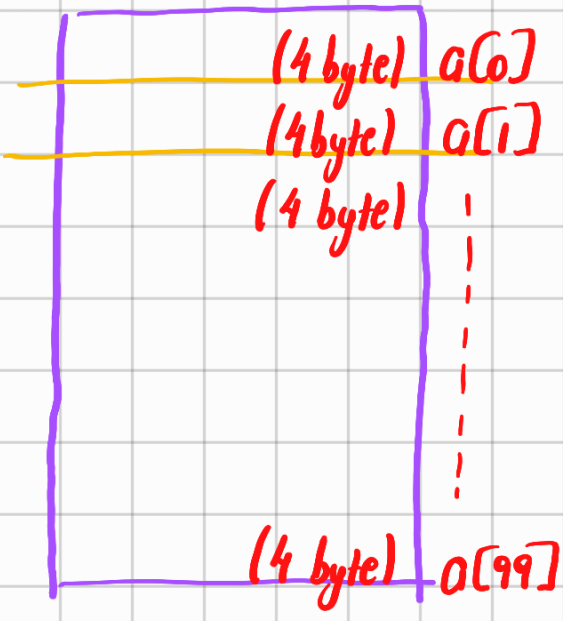
`int a[100];` → a[0] ____ a[99]

`float x[50];` → x[0] ____ x[49]

a[100]
x[-50]
x[100]
} segmentation fault

```
#define N 100

int a[N];
```




```
scanf("%d", &N);

  for(i=0; i<N; i++){

    int a[N];
    scanf("%d", &a[i]);

  }

  for(i=0; i<N; i++){

      printf("%f", x[i]);
```

C programming language

- general structure
  read / write operation
  control flow (if)
  Arithmetic operations, Comparision expressions
  scalar data type
  loops (for, while, do-while)
  Arrays
  Multi-dimensional Array

Array of characters (string)

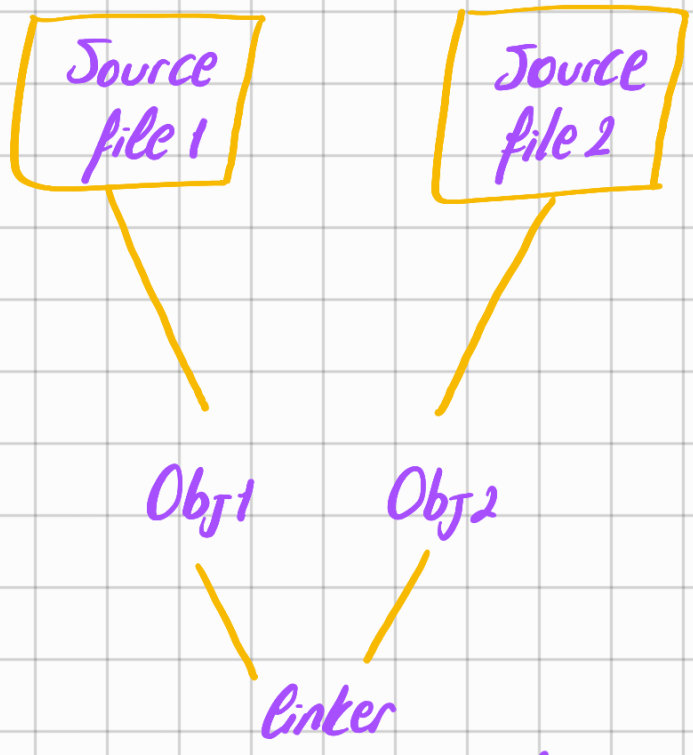
```
int main(){
   // comment line

   /* block 1
      block 2

   */
 return 0;

}
```

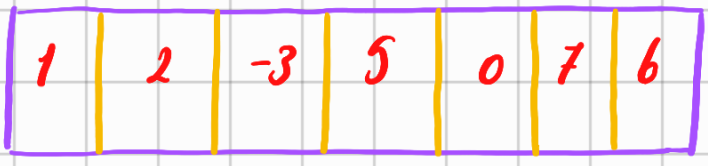
Ansi C-standart



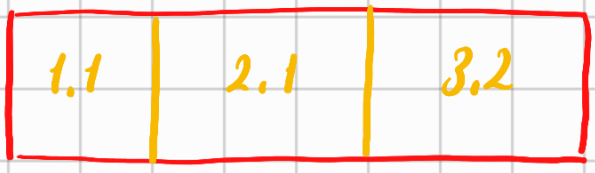
executable file

Read / write operations

decimal %d
character %c
floating-point %f
String %s
hexadecimal %x



Void type




```
Char a;
scanf("%c", &a);

printf("%c", a);

printf("%d", a);
```